OPTIMIZING DIETARY PLANS FOR ATHLETIC WEIGHT GAIN IN MALAYSIA: A LINEAR PROGRAMMING APPROACH

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1.0 INTRODUCTION

- **The Challenge:** Traditional manual diet planning for athletic weight gain is inefficient and economically challenging in Malaysia.
- **The Solution:** Applying Decision Science via a Linear Programming (LP) model to mathematically optimize a high-calorie, hypertrophy-focused diet.
- **The Impact:** Bridges the gap between strict sports nutrition and public health guidelines using affordable, local foods.

1.1 PROBLEM STATEMENT

- **Poor Practices:** Athletes lack accessible planning tools, often resorting to "dirty bulking" which causes excessive fat accumulation.
- **Inefficiency:** Manual calculation struggles to balance complex macronutrient targets against the cost of local Malaysian food.
- **The Consequence:** Athletes are left vulnerable to suboptimal performance, hidden nutritional deficiencies, and unnecessary financial strain.

1.2 RESEARCH OBJECTIVES

1. Determine daily macro/micronutrient requirements for athletic weight gain.
2. Formulate an LP model minimizing daily dietary costs using 10 local Malaysian foods.
3. Generate a cost-effective, constraint-compliant daily meal plan for muscle hypertrophy.

1.3 SIGNIFICANCE OF STUDY

- **Cost-Effective:** Provides a localized framework for hypercaloric surplus without financial burden.
- **Evidence-Based:** Replaces "dirty bulking" with a mathematically optimized, fat-minimizing diet.
- **Public Health:** Converts theoretical sports guidelines into accessible, real-world meal plans.

2.0 LITERATURE REVIEW

Article Title / Author(s)	Methodology	Objective	Main Findings
Sanchez & Larson-Meyer, 2025	Quantitative research Cross-sectional survey	Analyze athlete weight gain diets Focus on macro/supplement use	The majority of athletes rely on unguided, manual calorie increases, highlighting a critical need for structured dietary planning.
Everett, 2025	Narrative review	Assess adolescent dietary needs Identify nutritional risks	Establishes strict biological limits to prevent energy deficiency, specifically requiring 1.4–2.0 g/kg/day of protein while capping fat at 20–35%.
Amawi et al., 2024	Narrative review	Summarize evidence-based athletic guidelines	Confirms that athletic diets must be highly individualized with precise macronutrient parameters to ensure optimal muscle repair and growth.

3.0 METHODOLOGY

Research Design

- Utilizes a quantitative Linear Programming (LP) model to mathematically determine the most cost-effective athletic diet.
- Balances local Malaysian food prices against strict sports nutrition guidelines.

Decision Variables (x_j)

Defines the optimal daily servings of 10 specific local food items (j = 1, 2, ..., 10):

- Carbohydrates: Basmati Rice (x_1), Wholemeal Bread (x_2), Instant Oatmeal (x_3), Banana (x_4)
- Protein Foods: Chicken Breast (skinless) (x_5), Eggs (x_6), Ikan Kembung (x_7), Low-fat Milk (x_8), Yogurt (plain) (x_9)
- Micronutrient Sources: Spinach/Bayam (x_{10})

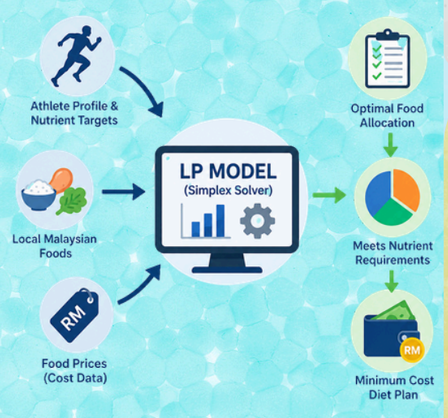
Objective Function

Designed to minimize the total daily financial cost (Z) while fulfilling all nutritional requirements.

Formula:

$$\text{Minimize } Z = \sum_{j=1}^{10} c_j x_j$$

Z = *Total daily cost of the diet (in RM)*
c_j = *Price per serving of food item j.*
x_j = *Number of servings of food item j..*



Nutritional & Biological Constraints

The model must strictly satisfy the following linear inequalities:

Macronutrients:

- **Energy:** Must meet or exceed daily caloric requirement (≥ Target).
- **Protein:** Must meet hypertrophy target (e.g., ≥ 1.6 g/kg of body weight).
- **Carbohydrates:** Must meet target for glycogen restoration (≥ Target).
- **Fat:** Must not exceed upper lipid limits (e.g., ≤ 30% of total calories).

Micronutrients:

- Must meet Recommended Nutrient Intakes (RNI) minimum thresholds for Iron, Calcium, Vitamin D, and Magnesium.

Feasibility Bounds:

- **Maximum Serving Limit:** $x_j \leq \text{Maximum Limit}$
- **Non-Negativity:** $x_j \geq 0 \quad \text{for all } j$

5.0 CONCLUSION AND RECOMMENDATIONS

Key Achievements

- **Cost-Optimized Hypertrophy:** The Linear Programming (LP) model successfully generated high-calorie diets that met all macronutrient targets, with the baseline athlete achieving a 2,760 kcal surplus for just RM 15.01/day.
- **High Algorithmic Adaptability:** The Simplex engine proved highly flexible across a 10-athlete cohort, instantly recalculating optimal diets to accommodate specific budgets, food allergies (e.g., eggs, ikan kembung), and taste aversions.
- **Biologically Realistic Outputs:** By enforcing maximum serving constraints from the Malaysian Dietary Guidelines 2020, the model produced practical, edible meal plans rather than mathematically extreme, unpalatable outputs.
- **Data-Driven Sports Nutrition:** The project successfully demonstrated that Decision Science can eliminate guesswork from athletic bulking, providing a highly personalized framework for students in Malaysia.

Study Limitations

- **The Vitamin D Bottleneck:** The model mathematically proved that achieving the 50 mcg Vitamin D requirement using only local whole foods is physically infeasible without violating public health maximum serving limits.
- **Static Market Pricing:** The objective function relied on static e-commerce prices (Shopee, 2025/2026), which dont track real-time inflation or supply chain fluctuations.
- **Data Standardization:** Integrating international nutrition data with Malaysian local metrics required extensive data cleaning.

4.0 RESULTS AND ANALYSIS

Athlete Cohort Profiles and Dietary Constraints (A - J):

Male Athlete	Current Bodyweight (kg)	Target Bodyweight (kg)	Daily Budget (RM)	Dietary Restrictions
A	60	70	35	None
B	62	66	35	Banana
C	54	62	30	Ikan Kembung
D	72	84	45	None
E	65	77	40	Banana
F	62	68	40	Wholemeal Bread
G	75	80	50	None
H	58	65	30	Banana
I	77	85	50	None
J	66	74	50	Eggs

The following formula calculates the total days required to safely reach the target body weight by applying a controlled daily energy surplus of 500 kcal.

$$Days = \frac{(TargetWeight - CurrentWeight) \times 7700}{500}$$

For Male Athlete A (60 kg to 70 kg, RM35/day budget, no dietary restrictions), the daily nutrition targets and 154-day hypertrophy plan were calculated as follows:

Male Athlete	Nutrient Required	Daily Recommend Intake	Daily Target Nutrition	Period (days)
A	Energy (kcal per Bodyweight (kg))	46	2760	154
	Protein (g per Bodyweight (kg))	2	120	
	Fat (g per Bodyweight (kg))	(Energy*0.3) ÷ 9	92	
	Carbohydrates (g per Bodyweight (kg))	6	360	
	Iron (mg per day)	10	10	
	Calcium (mg per day)	1200	1200	
	Vitamin D (mcg per day)	50	50	
	Magnesium (mg per Bodyweight (kg)	7	420	

The LP model for Athlete A was formulated using the following objective function and constraints:

Objective Function:

The algorithm's primary objective is to minimize the total daily cost (Z) in Ringgit Malaysia (RM):

$$\text{Minimize } Z = 0.79x_1 + 1.13x_2 + 1.07x_3 + 1.32x_4 + 1.75x_5 + 1.26x_6 + 1.72x_7 + 0.84x_8 + 3.92x_9 + 1.44x_{10}$$

Budget Constraint:

The minimized cost must not exceed the athlete's maximum daily budget:

$$0.79x_1 + 1.13x_2 + 1.07x_3 + 1.32x_4 + 1.75x_5 + 1.26x_6 + 1.72x_7 + 0.84x_8 + 3.92x_9 + 1.44x_{10} \leq 35.00$$

Macronutrient Constraints:

Energy (kcal):

$$362x_1 + 226x_2 + 379x_3 + 89x_4 + 120x_5 + 143x_6 + 158x_7 + 48x_8 + 129x_9 + 28x_{10} \geq 2760$$

Protein (g):

$$8.3x_1 + 13.4x_2 + 12.2x_3 + 1.09x_4 + 22x_5 + 13x_6 + 22x_7 + 3.4x_8 + 3.5x_9 + 2.91x_{10} \geq 120$$

Fat (g):

$$0.1x_1 + 3.3x_2 + 9x_3 + 0.33x_4 + 3x_5 + 10x_6 + 7.1x_7 + 1.3x_8 + 6.6x_9 + 0.6x_{10} \leq 92$$

Carbohydrates (g):

$$79.4x_1 + 36.9x_2 + 58x_3 + 22.8x_4 + 0x_5 + 1x_6 + 0x_7 + 5.7x_8 + 14x_9 + 2.64x_{10} \geq 360$$

Micronutrient Constraints:

Iron (mg):

$$1.5x_1 + 5.81x_2 + 3.3x_3 + 0.26x_4 + 0.4x_5 + 2x_6 + 1x_7 + 0x_8 + 0x_9 + 1.05x_{10} \geq 10$$

Calcium (mg):

$$10.2x_1 + 136x_2 + 0x_3 + 5x_4 + 5x_5 + 56x_6 + 20x_7 + 130x_8 + 0x_9 + 67x_{10} \geq 1200$$

Magnesium (mg):

$$0x_1 + 0x_2 + 100x_3 + 0x_4 + 28x_5 + 12x_6 + 0x_7 + 0x_8 + 0x_9 + 93x_{10} \geq 420$$

Biological Maximum Serving Constraints:

$$x_1 \leq 6,\ x_2 \leq 3.6,\ x_3 \leq 2.4,\ x_4 \leq 1.8,\ x_5 \leq 1.4,\ x_6 \leq 2,\ x_7 \leq 0.7,\ x_8 \leq 7.5,\ x_9 \leq 8.1$$

Non-Negativity Constraints:

$$x_j \geq 0 \quad \text{for all } j$$

Optimized Dietary Output for Athlete A

Type of Local Foods	Basmati Rice	Wholemeal Bread	Instant Oatmeal	Banana	Chicken Breast (skinless	Eggs	Ikan Kembung	Low-fat Milk	Yogurt (plain)	Spinach (Bayam)	Total Contributed	Target Constraint (A)
Maximum Servings (per 100g)	6	3.6	2.4	1.8	1.4	2	0.7	7.5	8.1	N/A		
Servings (per 100g)	2.0	3.6	2.4	0.0	0.0	0.5	0.0	4.1	0.0	1.9		
Price (RM per 100g)	0.79	1.13	1.07	1.32	1.75	1.26	1.72	0.84	3.92	1.44	15.01	35.00
Based on Standardized Serving Size (100g):												
Energy (kcal)	362	226	379	89	120	143	158	48	129	28	2760	2760
Protein (g)	8.3	13.4	12.2	1.09	22	13	22	3.4	3.5	2.91	120	120
Fat (g)	0.1	3.3	9	0.33	3	10	7.1	1.3	6.6	0.6	45	92
Carbohydrates (g)	79.4	36.9	58	22.8	0	1	0	5.7	14	2.64	457	360
Iron (mg)	1.5	5.81	3.3	0.26	0.4	2	1	0	0	1.05	35	10
Calcium (mg)	10.2	136	0	5	5	56	20	130	0	67	1200	1200
Vitamin D (mcg)	0	0	0	0	0	2	5	1.2	0	0	6	50
Magnesium (mg)	0	0	100	0	28	12	0	0	0	93	420	420

Summary of Cohort Nutritional Targets and Dietary Constraints

Athlete	A	B	C	D	E	F	G	H	I	J
Daily Budget (RM)	35	35	30	45	40	40	50	30	50	50
Dietary Restrictions	None	Banana	Ikan Kembung	None	Banana	Wholemeal Bread	None	Banana	None	Eggs
Energy (kcal per Bodyweight (kg))	2760	2852	2484	3312	2990	2852	3450	2668	3542	3036
Protein (g per Bodyweight (kg))	120	124	108	144	130	124	150	116	154	132
Fat (g per Bodyweight (kg))	92	95	83	110	100	95	115	89	118	101
Carbohydrates (g per Bodyweight (kg))	360	372	324	432	390	372	450	348	462	396
Iron (mg per day)	10	10	10	10	10	10	10	10	10	10
Calcium (mg per day)	1200	1200	1200	1200	1200	1200	1200	1200	1200	1200
Vitamin D (mcg per day)	50	50	50	50	50	50	50	50	50	50
Magnesium (mg per Bodyweight (kg)	420	434	378	504	455	434	525	406	539	462

Summary of Optimized Nutritional Achievements and Final Cost

Athlete	A	B	C	D	E	F	G	H	I	J
Optimized Cost (RM)	15.01	15.45	13.73	17.67	16.12	17.12	18.34	14.56	18.78	16.36
Energy (kcal per Bodyweight (kg))	2760	2852	2484	3312	2990	2852	3450	2668	3542	3036
Protein (g per Bodyweight (kg))	120	124	109	144	130	124	150	116	154	132
Fat (g per Bodyweight (kg))	45	47	40	57	50	54	59	43	60	42
Carbohydrates (g per Bodyweight (kg))	457	472	412	543	493	450	566	443	583	519
Iron (mg per day)	35	36	32	40	37	19	41	34	41	35
Calcium (mg per day)	1200	1200	1200	1200	1200	1200	1200	1200	1200	1200
Vitamin D (mcg per day)	6	6	6	7	7	13	7	6	7	5
Magnesium (mg per Bodyweight (kg)	420	434	378	504	455	434	525	406	539	462

Summary of Optimal Food Serving Allocations (per 100g)

Athlete	A	B	C	D	E	F	G	H	I	J
Basmati Rice	2.0	2.2	1.4	3.1	2.4	3.3	3.4	1.8	3.6	2.7
Wholemeal Bread	3.6	3.6	3.6	3.6	3.6	0.0	3.6	3.6	3.6	3.6
Instant Oatmeal	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Banana	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Chicken Breast (skinless)	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0	0.1	0.5
Eggs	0.5	0.7	0.0	1.7	1.0	2.0	2.0	0.3	2.0	0.0
Ikan Kembung	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Low-fat Milk	4.1	4.0	4.6	3.1	3.7	7.2	2.9	4.3	2.8	4.1
Yogurt (plain)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Spinach (Bayam)	1.9	2.0	1.5	2.6	2.2	1.7	2.8	1.7	2.9	2.2

CONTRIBUTION OF THE STUDY

- **Economic Framework:** Proves Malaysian athletes can achieve optimal, high-calorie hypertrophy diets for as low as RM 15.01 per day using local whole foods.
- **Scientific Proof:** The "Vitamin D Infeasibility" mathematically validates the necessity of supplementation, proving whole foods alone violate national portion limits.
- **Dynamic Personalization:** Delivers a scalable mathematical tool that instantly adapts to individual budgets, food allergies, and taste aversions.